

# Topological Memory.

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# Introduction

## Today:

- ▶ Proof.
  1. Walkthrough the math
  2. A
  3. B
- ▶ New experimental results.
- ▶ Answering the code teleportation.

# Our Code

## Code

In this session, we consider LDPC codes, where each bit adjoins to  $\Delta_1$  checks, and each check adjoins to  $\Delta_2$  bits. We denote by  $\Delta = \Delta_1 \Delta_2$ .

# Local Decoder

## definition

We say that  $D$  is a local decoder if when applied against  $2p(1 - p)$ -depolarized channel:

- ▶ There exists function  $M : (0, 1) \rightarrow (0, 1)$  such that the probability of each bit being flipped is  $M(2p(1 - p)) < p$ .
- ▶ There is a constant  $l$  such that for any bit, there are only  $(\Delta)^l$  bits on which the event of it being flipped depends.

For example, if we take  $k$  large enough and consider the  $(n, k)$ -Toric, then both the swift rule-and the flip upon majority are local decoders.

# Storing Logical State

## Logical State

For a random variable  $X$  we denote by:

1.  $\mathbf{E}_{\sim p}[X]$  - its expectation when measured after a single application of the  $p$ -depolarizing channel.
2.  $\mathbf{E}_{\sim q^{(t)}}[X]$  - its expectation when measured after  $t$  rounds, each initiated by a  $p$ -depolarized channel followed by an application of local decoder  $D$ .

# The Theorem

## Theorem

There is  $p_{\text{th}} \in (0, 1)$  such that for any  $p < p_{\text{th}}$  and any bounded function  $f$  with a supremum  $|f|$  it follows:

$$\mathbf{E}_{\sim q(t)}[f] \leq \mathbf{E}_{\sim p}[f] + t|f|e^{-\Theta(\sqrt{n})}$$

## Lemma

There exists a constant  $c > 0$ , such:

$$\mathbf{E}_{\sim 2p(1-p)}[f \circ D] \leq \mathbf{E}_{\sim p}[f \circ D] + |f|e^{-c\sqrt{n}}$$

# Theorem Proof

By induction on  $t$ .

$$\begin{aligned}\mathbf{E}_{\sim q^{(t)}}[f] &= \mathbf{E}_{\sim q^{(t-1)}}[\mathbf{E}_{\sim p}[f \circ D]] \\ &\leq \mathbf{E}_{\sim p}[\mathbf{E}_{\sim p}[f \circ D]] + (t-1)|\mathbf{E}_p[f \circ D]|e^{-c\sqrt{n}} \\ &\leq \mathbf{E}_{\sim 2p(1-p)}[f \circ D] + (t-1)|f|e^{-c\sqrt{n}} \\ &\leq \mathbf{E}_{\sim p}[f] + t|f|e^{-c\sqrt{n}}\end{aligned}$$

When in the last transition we used the lemma.

# Lemma Proof